

Jargon Buster

Root Mean Square (RMS): The sustained power in watts that an amplifier or speaker can output at any given time. It is a much more realistic measure of the sustained power of a speaker than PMPO.

Peak Music Power Output (PMPO): The maximum power that an amplifier can output over a short period of time. More of an advertising term nowadays, PMPO isn't an accurate depiction of a speaker's capabilities.

Frequency Response: The range of frequencies that the device is set to operate within.

Dolby Digital 5.1 (Dolby AC-3): Developed by Dolby, this system has five discrete sound channels in addition to Low Frequency Effects (LFE) that are directed to the subwoofer.

Dolby Pro Logic: An older standard, this packs in information for a centre and surround channel in the regular stereo channel. Essentially, there are no discrete left and right channels for the rear or surround speakers.

Dolby Pro Logic II: Audio encoded with Pro Logic II carries five channels of sound in a traditional two-channel stereo audio signal, which is then decoded back to five speakers using a Dolby Surround Pro Logic

II decoder. This technology can be used on any console that does not support Dolby Digital 5.1. The advantages of using it in game applications include minimal latency, full frequency range, compatibility with existing cables, and backward-compatibility with mono, stereo and legacy Dolby Surround Pro Logic equipment.

DTS: Short for Digital Theatre Systems, this is a standard invented by Steven Spielberg, and made its debut in *Jurassic Park*. It uses a sampling frequency of 96 kHz with 24 bits allocated to each of the six channels. Collector's edition DVD movies generally feature two soundtracks, featuring both Dolby 5.1 and DTS.

Dolby Digital EX and DTS ES: These are comparatively new standards that add an additional channel—the rear centre channel—and is implemented in 6.1 speakers. Both standards implement the rear channel in a similar way. For Dolby Digital EX, a special Dolby Pro Logic decoder is added to the surround channels, creating a rear centre channel. This decoding method first came to light with the theatrical release of *Star Wars Episode 1*. For DTS ES, an additional rear centre channel is supplemented to the two surround channels.

Dolby Pro Logic IIx: This is a spanking new technology developed by Dolby Laboratories and is the first and only technology to process native stereo or 5.1 signals into 6.1 or 7.1 channel output. Different modes such as Movie, Music and Game are available with this technology, with the Music mode featuring additional user controls such as Dolby centre width, Dolby panorama, and Dolby dimension.

THX: This is an abbreviation for "Tomlinson Holman's eXperiment." It's not a surround sound standard, but a seal of approval that is granted to audio equipment and theatres that meet its standards. This standard is further divided into THX Select (for installations up to 2,000 cubic feet) and THX Ultra (for between 2,000 and 3,000 cubic feet).

S/PDIF: Sony/Philips Digital Interface; a standard audio file transfer format. Developed jointly by Sony and Philips corporations, S/PDIF allows the transfer of digital audio signals from one device to another without having to be converted first to an analogue format. Maintaining the viability of a digital signal prevents the quality of the signal from degrading when it is converted to analogue.

Next, we played a DVD-Audio disc to test these speakers. The treble, vocals and bass were rated separately for this test. The Altec Lansing MX-5021 once again bettered every other set in this test. Be it vocals, treble or bass, the Altec Lansing MX-5021 proved itself to be the best 2.1 set as far as playing DVD audio was concerned. The Umax and the Techcom speakers were amongst the poorer performers in this test, and we do not recommend them for DVD audio.

Our music test suite comprised English and Hindi audio CDs and MP3 songs. Yet again, the Altec Lansing MX-5021, with its very high dynamic range, did not distort sound even at the maximum volume. The Corrs' folk song had vocals at an extreme range, and the Altec Lansing MX-5021 reproduced the sound more faithfully than did any other speakers in this test category.

The MX-5021 speakers also were capable of accurately reproducing the beats in the Enigma test song. The closest that could come to these speakers was the Altec Lansing CS21.

We moved on to listen to the Hindi audio CD song, and the MX-

5021 and CS21 were yet again leading the pack. With the MX-5021 and the Altec Lansing FX-6021, we could feel the *tabla* on the Zakir Hussain CD as if we were at a live concert.

MP3 music is what most people listen to these days, so this was an important test. The English MP3 songs played very nicely on the Altec Lansing MX-5021, FX-6021 and CS21. They accurately produced the different musical tones, instruments, as well as the vocals, with amazing clarity.

This was no different with the Hindi MP3 songs—these speakers soared ahead of the competition with the best scores.

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The Tech-Com SSD-801, Tech-Com SSD-901R, Umax PowerBeats UPB 980 and Zebronic ZEB-1400W performed below par in the music tests. These are entry-level speakers, and if you are an audiophile, keep away!

Having thus tested their mettle, we moved on to the synthetic tests.

We conducted frequency tests to gauge the capability of these speakers. The 50 Hz frequency was almost completely missed out by the Zebronic ZEB-SW8000, which means that this speaker set should not be expected to produce low bass tones. Many speakers such as the Umax, the two from Tech-Com, two from XFree, Zebronic ZEB-SW8000, Gigabyte GP-3A, Genius sw-i2.1, Artis S-700, Artis S-111R and the Frontech JIL-1811 performed very badly here and exhibited their weakness in the ability to reproduce a wide aural spectrum. The star in this test, again, was the Altec Lansing MX-5021, which trumped the competition comfortably. Only its brother, the Altec Lansing FX-6021, gave it some competition.

We then conducted tests to gauge the ability of the speakers to handle extreme bass and treble.